

A Crash Course in Arduino Programming

Lesson 2 Exercises

1. XOR evaluates as true when its inputs are different. Based on that, what would you propose XNOR does? Draw the truth tables for both.
2. Write a function that takes an integer as the input and returns the factorial of the input. For example: if you input 5, the function should compute $5! = 5*4*3*2*1$ and return 120. Can your function return the correct answer for zero factorial?

3. Write a while loop that prints every number between 0 and 100. Now write it as a do while loop. Now write it as a for loop.

4. Pick an example from today's lesson that used a button. If the example used an active high button, recreate the example with an active low button. If the example used an active low button, recreate the example with an active high button.

5. Create a function that prints the truth table for AND, OR, or NOT to the serial monitor.

6. Take any piece of example code that uses switch case and rewrite it to use if-else instead.

Challenge Problems

1. Take a four wheeled robot that uses a single H-bridge. Write functions for moving forward, moving backward, turning left, and turning right.
2. Set up 3 RGB LEDs and one regular LED. Use three buttons to the control the color of each LED. Have the regular LED turn on when the correct “RGB LED passcode” has been entered using the buttons.
3. Try writing a program that utilizes goto. Many programmers argue that the use of goto is always unnecessary and it should never be used. Do you agree? Explain.